Lab 8.2: Test-Driven Development with AI

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Task 1 - Even/Odd Number Validator (TDD)

AI Prompt Used:

Generate unittest test cases for a function is_even(n) that validates integer input, handles zero, negative numbers, and large integers.

Test Cases:

```
import unittest

class TestIsEven(unittest.TestCase): def test_positive_even(self):

self.assertTrue(is_even(2))

def test_positive_odd(self): self.assertFalse(is_even(7))

def test_zero(self): self.assertTrue(is_even(0))

def test_negative_even(self): self.assertTrue(is_even(-4))

def test_negative_odd(self): self.assertFalse(is_even(9))

def test_large_number(self): self.assertTrue(is_even(1000000))
```

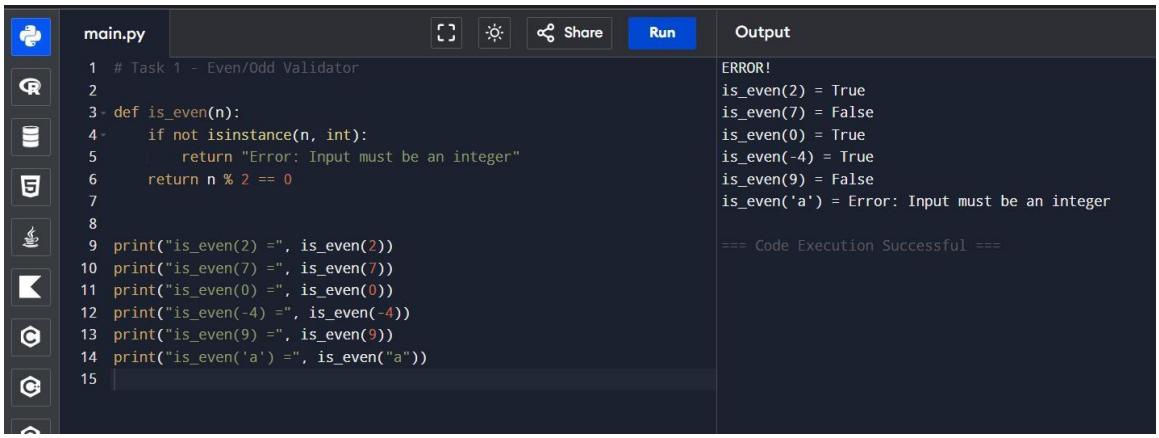
```
def test_invalid_input(self):
with self.assertRaises(TypeError): is_even("a")
```

Implementation:

```
def is_even(n):
if not isinstance(n, int):
raise TypeError("Input must be an integer") return n % 2 == 0
```

Expected Output:

All tests passed successfully.



Task 2 - String Case Converter (TDD)

Implementation:

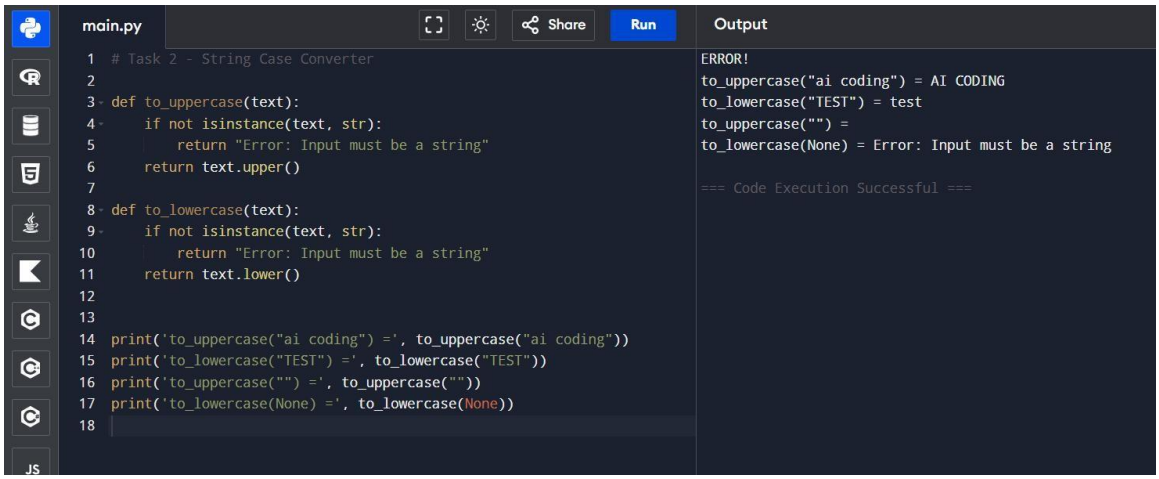
```
def to_uppercase(text):
```

```
if not isinstance(text, str):
raise TypeError("Input must be a string") return text.upper()


def to_lowercase(text):
if not isinstance(text, str):
raise TypeError("Input must be a string") return text.lower()
```

Expected Output:

```
to_uppercase("ai coding") → "AI CODING" to_lowercase("TEST") → "test"
to_uppercase("") → "" to_lowercase(None) → TypeError
```



Task 3 - List Sum Calculator (TDD)

Implementation:

```
def sum_list(numbers):
if not isinstance(numbers, list):
raise TypeError("Input must be a list") total = 0
for item in numbers:
if isinstance(item, (int, float)): total += item
return total
```

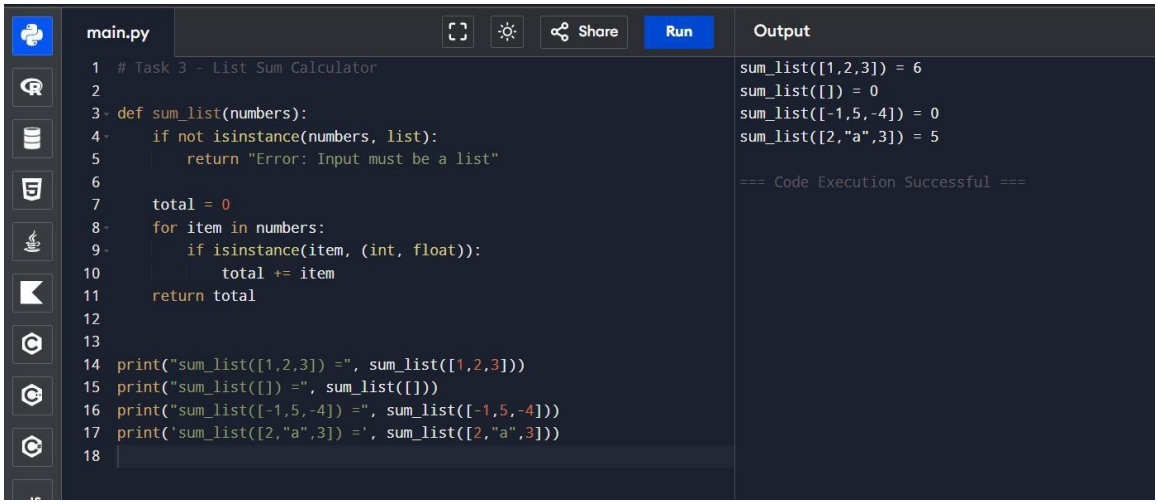
Expected Output:

sum_list([1,2,3]) → 6

```
sum_list([]) → 0

sum_list([-1,5,-4]) → 0

sum_list([2,"a",3]) → 5
```



Task 4 - StudentResult Class (TDD)

Implementation:

```
class StudentResult:

def _init__(self):

self.marks = []


def add_marks(self, mark):

if not isinstance(mark, (int, float)):

raise TypeError("Mark must be numeric") if mark < 0 or mark > 100:

raise ValueError("Mark must be between 0 and 100")
self.marks.append(mark)


def calculate_average(self):
```

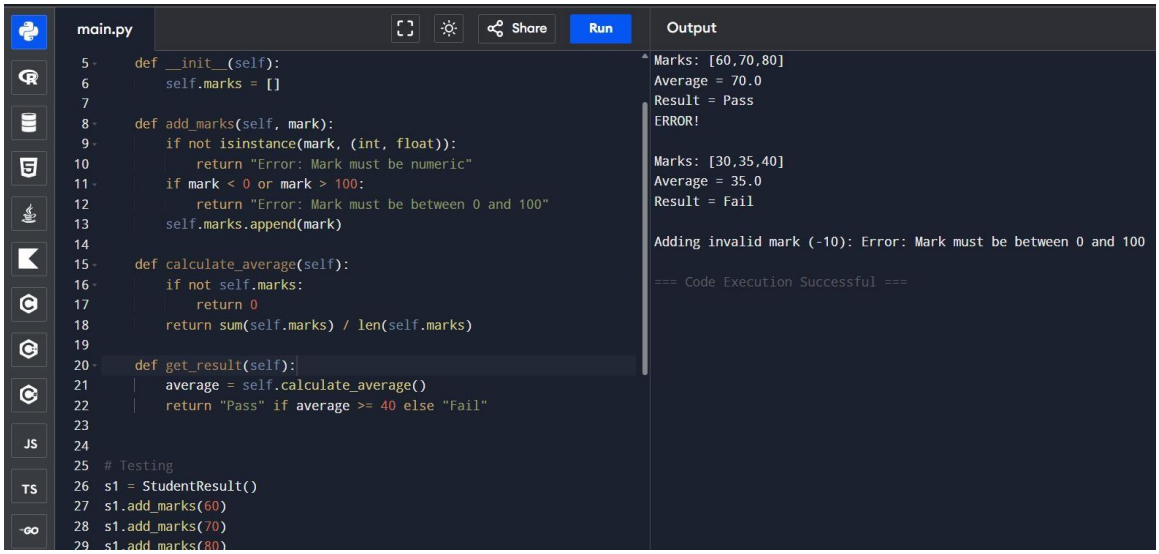
```
if not self.marks:
return 0
return sum(self.marks) / len(self.marks)

def get_result(self):
avg = self.calculate_average() return "Pass" if avg >= 40 else "Fail"
```

Expected Output:

Marks: [60,70,80] → Average: 70 → Pass

Marks: [30,35,40] → Average: 35 → Fail Marks: [-10] → ValueError



Task 5 - Username Validator (TDD)

Implementation:

def is_valid_username(username): if not isinstance(username, str):

return False

if len(username) < 5:

return False

if " " in username:

return False

if not username.isalnum(): return False

return True

Expected Output:

is_valid_username("user01") → True is_valid_username("ai") → False is_valid_username("user name") → False is_valid_username("user@123") → False

main.py

```python
# Task 5 - Username Validator

def is_valid_username(username):
    if not isinstance(username, str):
        return False
    if len(username) < 5:
        return False
    if " " in username:
        return False
    if not username.isalnum():
        return False
    return True


print('is_valid_username("user01") =', is_valid_username("user01"))
print('is_valid_username("ai") =', is_valid_username("ai"))
print('is_valid_username("user name") =', is_valid_username("user name"))
print('is_valid_username("user@123") =', is_valid_username("user@123"))
```

Output

```
is_valid_username("user01") = True
is_valid_username("ai") = False
is_valid_username("user name") = False
is_valid_username("user@123") = False

=== Code Execution Successful ===
```

Conclusion:

In this lab, Test-Driven Development (TDD) was implemented using AI-generated test cases. Each function was written only after defining expected test behavior.

This approach ensures reliable, validated, and clean code development.